SOLUTION
SECTION-I

Single Correct Answer Type **9 Q. [3 M (–1)]**

1. Ans. (D)

Sol. In this case reduced mass $m' = \frac{mm}{m+m} = \frac{m}{2}$

So $E_n' = \frac{m'}{m} = \frac{E_1}{n^2} = \frac{E_1}{2n^2}$

Here E_1 is energy of electron in first orbit in ordinary hydrogen atom.

2. Ans. (A)

Sol. For lyman alpha

$$\frac{hc}{\lambda} = RZ^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right] \Rightarrow \frac{hc}{1215} = R\left[\frac{1}{1^2} - \frac{1}{2^2}\right]$$

$$\Rightarrow \frac{hc}{R} = 1215 \times \frac{3}{4} = 911.25 \text{ Å}$$

To photo–ionization of hydrogen atoms in ground state, the minimum wavelength

$$\frac{hc}{\lambda_{min}} = R\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right] \Rightarrow \lambda_{min} = \frac{hc}{R} = 911.25\text{Å}$$

To photo–ionization of hydrogen atoms in first excited state, the minimum wavelength

$$\frac{hc}{\lambda_{min}} = R\left[\frac{1}{2^2} - \frac{1}{\infty^2}\right] \Rightarrow \lambda_{min} = \frac{4hc}{R} = 3645\text{Å}$$

To photo–ionization of He^+ atoms in ground state, the minimum wavelength

$$\frac{hc}{\lambda_{min}} = R(2)^2\left[\frac{1}{1^2} - \frac{1}{\infty^2}\right] \Rightarrow \lambda_{min} = \frac{hc}{4R} = 227.8\text{Å}$$

To photo–ionization of He^+ atoms in first excited state, the minimum wavelength

$$\frac{hc}{\lambda_{min}} = R(2)^2\left[\frac{1}{2^2} - \frac{1}{\infty^2}\right] \Rightarrow \lambda_{min} = \frac{hc}{R} = 911.25\text{Å}$$

3. Ans. (D)

Sol. $E = a(1 + \cos 2\pi f_1 t)\cos 2\pi f_2 t = a\cos 2\pi f_2 t + a\cos 2\pi f_1 t \cos 2\pi f_2 t$

$$\Rightarrow E = a\cos 2\pi f_2 t + \frac{1}{2}a\cos 2\pi(f_1 + f_2)t + \frac{1}{2}a\cos 2\pi(f_1 - f_2)t$$

This is a complex vibration consisting of harmonic vibrations of frequencies f_2, $(f_1 + f_2)$ and $(f_1 - f_2)$
The highest is $(f_1 + f_2)$. So, $h\nu = \phi + T_{max}$

$$T_{max} = h(f_1+f_2) - \phi = \frac{6.6 \times 10^{-34}}{1.6 \times 10^{-19}}\left(3.6 \times 10^{15} + 1.2 \times 10^{15}\right) - 2.35 = 17.45 \text{ eV}$$

4. Ans. (A)

Sol. $p = \frac{h}{\lambda}$

$$K.E. = \frac{p^2}{2m} = \frac{h^2}{2m\lambda^2}$$

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If entire K.E. of electron is converted into photon then

$$\frac{h^2}{2m\lambda^2} = \frac{hc}{\lambda_0}$$

$$\lambda_0 = \frac{2mc\lambda^2}{h}$$

5. Ans. (D)

Sol. Transition from n = 3 to 2 (first line OF Balmer Series corresponds to longest wavelength), photon emitted with momentum

$$P_{photon} = \frac{h}{\lambda} = Rh\left(\frac{1}{3^2} - \frac{1}{2^2}\right) = \frac{5Rh}{36}$$

Apply Momentum Conservation

$$P_{photon} = P_{atom}$$

$$\frac{5Rh}{36} = MV$$

$$V = \frac{5Rh}{36M}$$

6. Ans. (D)

Sol. From Moseley's law, $\sqrt{\nu} = a(Z-b)$

$$\frac{1}{\lambda} \propto k(Z-b)^2 \qquad \{k \text{ is a constant}\}$$

For K_α : b = 1

$$\frac{\frac{1}{\lambda_{K_\alpha \text{ copper}}}}{\frac{1}{\lambda_{K_\alpha \text{ impurity}}}} = \left(\frac{Z_{Cu}-b}{Z_{imp}-b}\right)^2 \Rightarrow \frac{\lambda_{K_\alpha \text{ impurity}}}{\lambda_{K_\alpha \text{ copper}}} = \left(\frac{Z_{Cu}-b}{Z_{imp}-b}\right)^2 \qquad \text{so } Z_{imp} = 26$$

7. Ans. (B)

Sol. $\lambda_{min} = \frac{hc}{eV}$

Cut off wavelength depends on the energy of the accelerated electrons and is independent of nature of target.

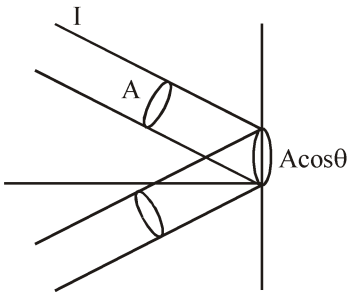
$$\lambda_{K\alpha} \propto \frac{1}{(z-b)^2}$$

characteristic wavelength depend on atomic no and cut off wavelength depend on energy of e^-.

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8. Ans. (A)

Sol.



$$\left(\frac{dp}{dt}\right)_{incident} = \frac{I}{C} A\cos^2\theta$$

$$\left(\frac{dp}{dt}\right)_{reflected} = \frac{-I}{C} A\, 0.5\cos^2\theta$$

$$F = \frac{3}{2}\frac{I}{C} A\cos^2\theta$$

$$\frac{F}{A} = P = \frac{3I}{2C}\cos^2\theta = \frac{3I}{4C}$$

9. Ans. (A)

Sol. According to photo electric effect equation :

$$KE_{max} = \frac{hc}{\lambda} - \phi_0$$

$$\frac{p^2}{2m} = \frac{hc}{\lambda} - \phi_0$$

$$\frac{(h/\lambda_d)^2}{2m} = \frac{hc}{\lambda} - \phi_0$$

Assuming small changes, differentiating both sides,

$$\frac{h^2}{2m}\left(-\frac{2d\lambda_d}{\lambda_d^3}\right) = -\frac{hc}{\lambda^2} d\lambda$$

$$\frac{d\lambda_d}{d\lambda} \propto \frac{\lambda_d^3}{\lambda^2}$$

Multiple Correct Answer Type **7 Q. [4 M (–1)]**

10. Ans. (A, C, D)

Sol. $E = \frac{1240}{248}$ eV

$E_1 = 5 - 2.2 = 2.8$ eV
$E_2 = 5 \times 0.8 - 2.2 = 1.8$ eV
$E_3 = 5 \times 0.8^2 - 2.2 = 1$ eV
$E_4 = 5 \times 0.8^3 - 2.5 = 0.36$ eV

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11. Ans. (A,B,D)

Sol. $\Delta E = E_2 - E_1 = 13.6\ Z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right](eV)$

for K_α : $n_1 = 1, n_2 = 2$
for K_β : $n_1 = 1, n_2 = 3$
value of Z is 3 (Li), 4(Be), 11(Na), 12(Mg), 13(Al)

12. Ans. (A,B,D)

Sol. $E_{excitation} = \frac{1}{2}\frac{m}{2}u^2 = E_2 - E_1 = \frac{E_0}{2}$

$$\lambda_0 = \frac{hc}{\sqrt{2m_e\left[(E_2 - E_1) - \phi\right]}}$$

$$= \frac{hc}{\sqrt{2m_e\left[\frac{E_0}{2} - \phi\right]}} = \frac{hc}{\sqrt{m_e E_0 - 2\sigma m_e}}$$

Also $\Rightarrow$ Energy transferred

$$\Rightarrow \frac{1}{2}mu^2 - \frac{1}{2}m\left(\frac{u}{2}\right)^2 = mu^2\left(\frac{1}{2} - \frac{1}{8}\right) = mu^2\left(\frac{3}{8}\right) = \frac{mu^2}{4}, \frac{3}{2} = \frac{E_0}{2}.\frac{3}{2} = \frac{3E_0}{4}$$

Also For second case,

$$E_{excitation} = \frac{1}{2}\frac{4m}{5}u'^2 = E_2' - E_1' = 4(E_2 - E_1) = \frac{4}{5}E_0'$$

$$E_0' = 5(E_2 - E_1) = \frac{5E_0}{2}z$$

13. Ans. (A, D)

Sol. $\lambda = \frac{h}{mv}$

$$\frac{dh}{dt} = -\frac{h}{mv^2}\frac{dV}{dt}$$

$$10^{-4} = -\frac{10^{-34} \times 6.6}{6.6 \times 10^{-30}}\frac{1}{V^2}\frac{dV}{dt}$$

$$\frac{dV}{dt} = -V^2$$

$$\int_{10}^{V}\frac{dV}{V^2} = -\int_0^t dt$$

$$\frac{1}{V} - \frac{1}{10} = t$$

$$\frac{1}{V} = \frac{1 + 10t}{10}$$

$$V = \frac{10}{1 + 10t}$$

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at t = 0.9 s V = 1 m/s

Also $a = -V^2$

$|a| = 1\ m/s^2$

14. Ans. (A,B)

Sol. For Balmer series, electron jumps to n = 2 from higher energy level. Since for n = 1, energy is minimum, so the e^- in n = 2 jumps to n = 1 thus giving a Lyman photon having energy of 10.2 eV or wavelength

$$\frac{hc}{E} \approx 1225\ \text{Å}$$

15. Ans. (A,D)

Sol. $\lambda_{min} = \frac{hc}{eV_0}$

If V_0 increases, λ_{min} reduces

$\therefore$ more photons are emitted

$\therefore$ Intensity increases.

16. Ans. (A, B, D)

Sol. As radius $r \propto \frac{n^2}{z}$

$$\Rightarrow \frac{\Delta r}{r} = \frac{\left(\frac{n+1}{z}\right)^2 - \left(\frac{n}{z}\right)^2}{\left(\frac{n}{z}\right)^2} = \frac{2n+1}{n^2} \approx \frac{2}{n} \propto \frac{1}{n}$$

as energy $E \propto \frac{z^2}{n^2}$

$$\Rightarrow \frac{\Delta E}{E} = \frac{\frac{z^2}{n^2} - \frac{z^2}{(n+1)^2}}{\frac{z^2}{(n+1)^2}} = \frac{(n+1)^2 - n^2}{n^2.(n+1)^2}.(n+1)^2 \Rightarrow \frac{\Delta E}{E} = \frac{2n+1}{n^2} \simeq \frac{2n}{n^2} \propto \frac{1}{n}$$

as angular momentum $L = \frac{nh}{2\pi}$

$$\Rightarrow \frac{\Delta L}{L} = \frac{\frac{(n+1)h}{2\pi} - \frac{nh}{2\pi}}{\frac{nh}{2\pi}} = \frac{1}{n} \propto \frac{1}{n}$$

$$1.6\times10^{-19}\times2 = \frac{h\times3\times10^8}{3000\times10^{-10}} - \frac{h\times3\times10^8}{4000\times10^{-10}} + 1.6\times10^{-19}$$

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$$1.6\times10^{-19}=\frac{h\times3\times10^{8}}{10^{-7}}\left(\frac{1}{3}-\frac{1}{4}\right)=\frac{h\times3\times10^{8}}{10^{-7}}\left[\frac{4-3}{12}\right]$$

$$1.6\times10^{-19}=\frac{h\times3\times10^{8}}{10^{-7}}\times\frac{1}{12}$$

$$1.6\times4\times\frac{10^{-19}\times10^{-7}}{10^{8}}=h$$

6.4×10^{-34} Js = h

Linked Comprehension Type (2 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

17. Ans. (A)

Sol. UV energy flux at a distance of 1m $=\frac{0.1}{4\pi\times1^2}$

cross section (effective area) of atom = $\pi\times(2\times10^{-10})^2=4\pi\times10^{-20}$ m^2
Energy required to eject a photoelectron from potassium = 2.2 eV $\equiv 2.2\times1.6\times10^{-19}$ J.

$\Rightarrow$ Exposure time = $\frac{2.2\times1.6\times10^{-19}}{\left(\frac{0.1}{4\pi\times1^2}\right)\left(4\pi\times10^{-20}\right)}=352$ s

18. Ans. (A)

Sol. Flux of photon at the cathode = $\left(\frac{0.1}{4\pi\times1^2}\right)\left(\frac{1}{\text{photon energy}}\right)$ = 1.015×10^{16} photons/ sec m^2

Saturation current = (photon flux × effective area of cathode) × 5/100 × $1.6\times10^{-19}=3.25\times10^{-8}$ A.

19. Ans. (B)

Sol. Cut off potential $=\frac{(4.897-2.22)\text{eV}}{e}=2.69$ volts.

20. Ans. (B)

Sol. $\frac{hc}{\lambda_0}=\phi_{wf}\Rightarrow\lambda_0=6200\text{Å}$

for emission $\lambda<\lambda_0\Rightarrow 3000+40t=6200\Rightarrow t=80$ sec

21. Ans. (B)

Sol. $\frac{P}{100}=\frac{dN}{dt}\frac{hc}{\lambda}$ for 3 minute $\Rightarrow\frac{dN}{dt}=\frac{\lambda}{hc}\times1\Rightarrow dN=\int_0^{80}\frac{\lambda}{hc}dt$; $N'=\frac{3.71\times10^{21}}{20}$

for one hour $N = N'\times20 = 3.71\times10^{21}$

22. Ans. (B)

Sol. $\left(\frac{dN}{dt}\right)=\frac{P\lambda}{hc}=\frac{100}{hc}(3000+40t)$

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SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **5 Q. [4 M (0)]**

23. Ans. 2

Sol. K = U + K

$$\frac{p_1^2}{2m} = K_1 = 16eV$$

$$\frac{p_2^2}{2m} = K_2 = 4eV$$

$$\lambda_1^2 \times 16 = \lambda_2^2 \times 4$$

$$2 = \frac{\lambda_2}{\lambda_1}$$

24. Ans. 8

Sol. Given distance between electrode = 4×10^{-2}m

Average distance travelled by electrons between successive collision = 8×10^{-5} m.

Thus, $\frac{eV}{4\times10^{-2}} \times 8\times10^{-5} = 16eV \Rightarrow V = 8keV$

25. Ans. 8

Sol. As we know,

$$F = qvB = m\frac{v^2}{R} \Rightarrow v = \frac{q}{m}BR$$

The kinetic energy of the photoelectron

$$\frac{1}{2}mv^2 = \frac{1}{2}\frac{e^2B^2R^2}{m}$$

$$= \frac{1}{2}\frac{\left(1.6\times10^{-19}\right)^2\left(2\times10^{-2}\right)^2\left(23\times10^{-3}\right)^2}{\left(9.1\times10^{-31}\right)}$$

$= 2.97 \times 10^{-15}$ J

$$= \frac{2.97\times10^{-15}}{1.6\times10^{-19}} = 18.4\,KeV$$

Energy of the incident photon = $\frac{hc}{\lambda}$

$$= \frac{12.4}{0.50} = 24.8keV$$

Therefore binding energy = 24.8 – 18.4 = 6.4 KeV = 6400 eV

26. Ans. 3

Sol. Atom will exist if size of nucleus is less than the orbit of π-meson

$$\therefore 1.6\times10^{-15}Z^{1/3} < \frac{0.53}{265Z} \Rightarrow Z < 37.5$$

$$r = \frac{n^2h}{4\pi^2Kzem_\pi}$$

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$r = \frac{\varepsilon_0 h^2}{\pi m \pi e^2} \times \frac{n^2}{z} \Rightarrow (1^{st} \text{ orbit})$ $\qquad r = \frac{0.53}{265} \times \frac{1}{Z}$

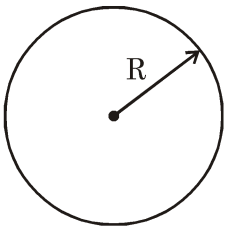
Atom will exist if size of nucleus is less than the orbit of π-meson

$\therefore\ 1.6 \times 10^{-15}\, Z^{1/3} < \frac{0.53}{265Z} \Rightarrow Z < 37.5$

$\therefore\ Z_{max} = 37$

27. Ans. 7

Sol.



$R = 1$ cm

$\phi = 4.7$ eV

$\frac{hc}{\lambda} = \phi - eV$

$\frac{1240(ev)(nm)}{200(nm)} = 4.7(eV) + eV$

$\frac{1240}{200} e = 4.7eV + eV$

$6.2 - 4.7 = V$ $\qquad \therefore\ V = 1.5$ volt

$\frac{1}{4\pi \in_0} \frac{Q}{R} = 1.5$

$(9 \times 10^9) \frac{Ne}{\frac{1}{100}} = 1.5$

$9 \times 10^{11}\, Ne = 1.5;$ $\qquad N = \frac{1.5}{9 \times 10^{11} \times 1.6 \times 10^{-19}} = \frac{15}{16} \times \frac{1}{9} \times 10^8$

$= \frac{5}{3 \times 16} \times 10^8 = \frac{50}{48} \times 10^7$

$Z = 7$

Subjective Type **2 Q. [4 M (0)]**

28. Ans. 1.1×10^{12}

Sol. $\lambda_1 = 4144$ Å

$E_1 = \frac{12400}{4144} eV$, $E_1 = 3$ eV

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$\lambda_2 = 4972$ Å

$$E_2 = \frac{12400}{4972} = 2.5\,eV$$

$\lambda_3 = 6216$

$$E_3 = \frac{12400}{6216} = 2\,eV$$

number of photo electron emitted in 2 second

$$= 2\times\left(\frac{1.2\times10^{-3}}{3\times1.6\times10^{-19}} + \frac{1.2\times10^{-3}}{2.5\times1.6\times10^{-19}}\right)\times10^{-4}$$

$$= 2\times10^{16}\left(\frac{1.2}{4.8} + \frac{1.2}{400}\right)\times10^{-4}$$

$= 1.1 \times 10^{12}$

29. **Ans.** KE $\cong$ 151 eV, d_{least} = 0.5 Å

Sol. $\therefore$ if d is multiple of 0.5 Å then standing waves would be fomred

$$\text{Energy} = \frac{p^2}{2M}$$

$$= \frac{h^2}{2\lambda^2 m_\alpha}$$

$$= \frac{\left(6.63\times10^{-34}\right)^2}{2\times\left(0.5\times10^{-10}\right)^2\times9.11\times10^{-31}}$$

$$= \frac{9.65\times10^{-17}}{4} = 2.412\times10^{-17} = 151\,eV$$

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